Recoil and explosions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart has momentum -15 kg·m/s and mass 3 kg. Find velocity and its new momentum if velocity reverses at unchanged speed.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part's momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A 1 kg cart leaves at +4 m/s; a 2 kg cart must have momentum -4 kg·m/s, hence velocity -2 m/s.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two carts push apart from rest on an ideal track. Draw their momentum arrows. How must the two momenta compare?



A moves left B moves right

Marzano check - circle: ① ② ③

04 Apply the physics

An astronaut of mass 60 kg and a 3 kg tool are initially at rest relative to a spacecraft. The astronaut pushes the tool so it moves at +8 m/s in that frame. Label the model and find the astronaut's velocity. Neglect external impulse.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A projectile's velocity is (12, -16) m/s. Find its speed. What information would locate it?
Label separate horizontal and vertical motion.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two fragments of masses 2 and 3 kg separate from rest. The 2 kg fragment moves at +9 m/s. Find the other velocity and total kinetic energy. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 60 kg person and 10 kg board start at rest. After a push, the board moves at +3 m/s relative to the ground. Find the person's ground velocity and the board's speed relative to the person. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3